EXPERIMENT : 2	ROLL NO	MANIPULATION OF FILE MANAGEMENT TASKS IN HADOOP
DATE:	23DI16	

Aim:

To install Hadoop and Monitor setup using web-based tool

COMPONENTS REQUIRED:

S.no	FACILITIES REQUIRED	QUANTITY
1	System	01
2	Operating System	Ubuntu

Procedure:

Step 1: Load hdfs commands on ubuntu terminal

$~ sudo apt update

```
dit@dit-OptiPlex-3060:~$ sudo apt update
[sudo] password for dit:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,820 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [916 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,084 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [733 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [408 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [54.5 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [13.9 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [4,726 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [476 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [112 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.0 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe i386 Packages [788 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,240 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [308 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [359 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [29.8 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:23 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [7,152 B]
Get:24 http://in.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 DEP-11 Metadata [212
B]
Get:25 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [9,696
B]
Get:26 http://in.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 DEP-11 Metadata [212
```

$~ hdfs dfs -command

```
hdoop@dit-OptiPlex-3060:~$ hdfs dfs -command
-command: Unknown command
Usage: hadoop fs [generic options]
        [-appendToFile [-n] <localsrc> ... <dst>]
        [-cat [-ignoreCrc] <src> ...]
        [-checksum [-v] <src> ...]
        [-chgrp [-R] GROUP PATH...]
        [-chmod [-R] <MODE[,MODE]... | OCTALMODE> PATH...]
        [-chown [-R] [OWNER][:[GROUP]] PATH...]
        [-concat <target path> <src path> <src path> ...]
        [-copyFromLocal [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
        [-copyToLocal [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
        [-count [-q] [-h] [-v] [-t [<storage type>]] [-u] [-x] [-e] [-s] <path> ...]
        [-cp [-f] [-p | -p[topax]] [-d] [-t <thread count>] [-q <thread pool queue size>] <src> ... <dst>]
        [-createSnapshot <snapshotDir> [<snapshotName>]]
        [-deleteSnapshot <snapshotDir> <snapshotName>]
        [-df [-h] [<path> ...]]
        [-du [-s] [-h] [-v] [-x] <path> ...]
        [-expunge [-immediate] [-fs <path>]]
        [-find <path> ... <expression> ...]
        [-get [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
        [-getfacl [-R] <path>]
        [-getfattr [-R] {-n name | -d} [-e en] <path>]
        [-getmerge [-nl] [-skip-empty-file] <src> <localdst>]
        [-help [cmd ...]]
        [-ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]]
        [-mkdir [-p] <path> ...]
        [-moveFromLocal [-f] [-p] [-l] [-d] <localsrc> ... <dst>]
        [-moveToLocal <src> <localdst>]
        [-mv <src> ... <dst>]
        [-put [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
        [-renameSnapshot <snapshotDir> <oldName> <newName>]
        [-rm [-f] [-r|-R] [-skipTrash] [-safely] <src> ...]
        [-rmdir [--ignore-fail-on-non-empty] <dir> ...]
        [-setfacl [-R] [{-b|-k} {-m|-x <acl_spec>} <path>]|[--set <acl_spec> <path>]]
        [-setfattr {-n name [-v value] | -x name} <path>]
        [-setrep [-R] [-w] <rep> <path> ...]
        [-stat [format] <path> ...]
        [-tail [-f] [-s <sleep interval>] <file>]
        [-test -[defsrwz] <path>]
        [-text [-ignoreCrc] <src> ...]
        [-touch [-a] [-m] [-t TIMESTAMP (yyyyMMdd:HHmmss) ] [-c] <path> ...]
        [-touchz <path> ...]
        [-truncate [-w] <length> <path> ...]
```

Step 2: Create new directory using mkdir command

$~ hdfs dfs -mkdir /path/to/directory

```
hdoop@dit-OptiPlex-3060:~$ hdfs dfs -mkdir /Downloads
hdoop@dit-OptiPlex-3060:~$ hdfs dfs -ls
```

Step 3: View the created directory

$~ hdfs dfs -ls

```
hdoop@dit-OptiPlex-3060:~$ hdfs dfs -ls /
Found 2 items
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 14:03 /Downloads
```

Step 4: To delete the director

$~ hdfs dfs -rm -r /path/to/directory

```
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -rm -r /ravi
Deleted /ravi
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -ls /
Found 3 items
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 15:11 /Aagash
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 14:03 /Downloads
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 15:15 /dwarakesh
hdoop@dit-OptiPlex-3060:/home/dit$
```

Step 5: To move file from one directory to another

$~ hdfs dfs -mv /source/path /destination/path

```
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -mkdir /dwarakesh/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -mv /dwarakesh/dwara.txt /Aagash/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -ls /dwarakesh
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -ls /Aagash
Found 1 items
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 15:20 /Aagash/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$
```

Step 6: To copy file from one directory to another

$~ hdfs dfs -cp /source/path /destination/path

```
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -cp /Aagash/dwara.txt /dwarakesh/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -ls /Aagash
Found 1 items
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 15:20 /Aagash/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$ hdfs dfs -ls /dwarakesh
Found 1 items
drwxr-xr-x   - hdoop supergroup          0 2025-12-09 15:25 /dwarakesh/dwara.txt
hdoop@dit-OptiPlex-3060:/home/dit$
```

Result :

Thus we have Successfully managed Hadoop commands to manage files on hdfs